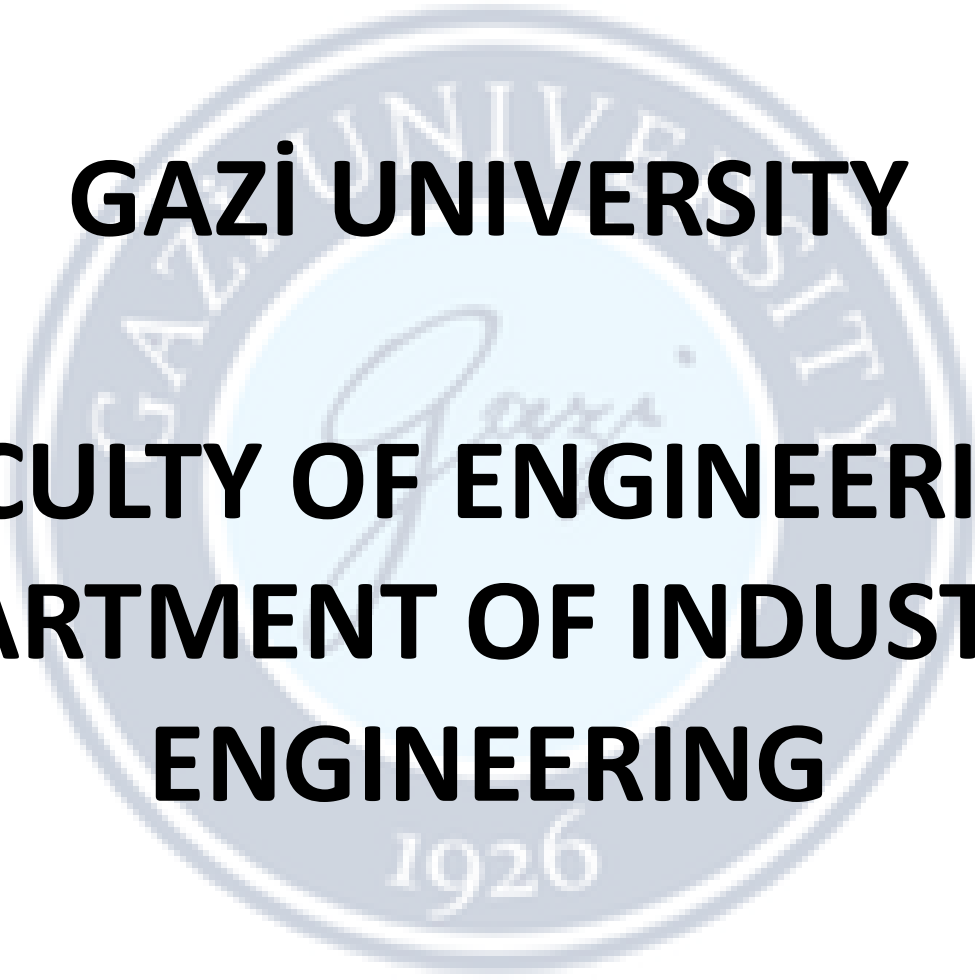


The background features a large, light blue circular seal of Gazi University. The seal contains the university's name in Turkish and English, the year 1926, and a central emblem with a signature.

GAZİ UNIVERSITY

FACULTY OF ENGINEERING

DEPARTMENT OF INDUSTRIAL ENGINEERING

Lecturer : Dr Ercan Ezin

IE326-DATABASE MANAGEMENT SYSTEMS

WEEK 14:EXAMPLE DB DESIGN/AI-AIDED DESIGN

PHYSICAL DESIGN EXAMPLE



PHYSICAL DESIGN

ER DIAGRAM

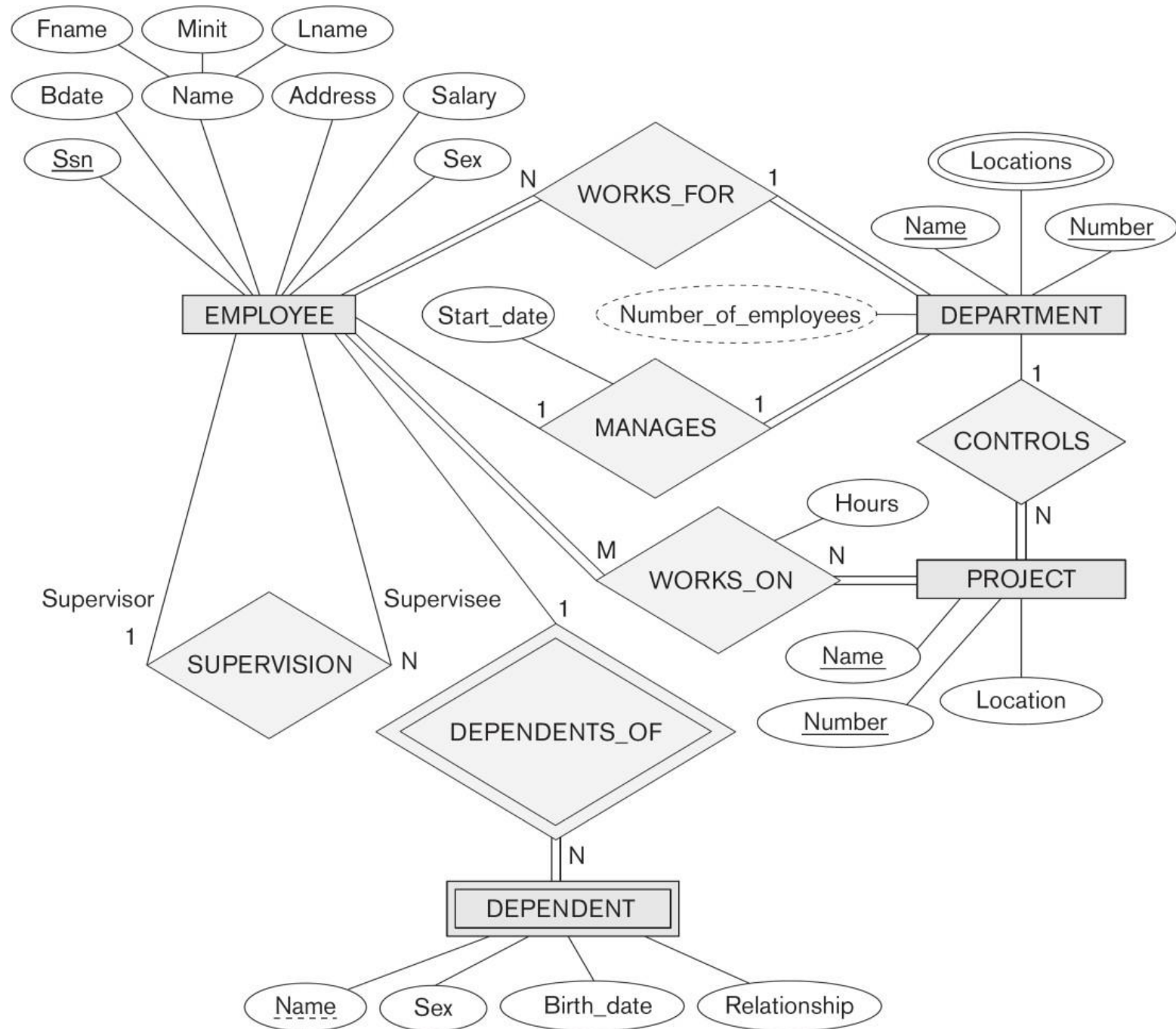


Table Creation

```
-- =====  
-- DROP TABLES  
-- =====  
  
DROP TABLE IF EXISTS works_on CASCADE;  
DROP TABLE IF EXISTS dependent CASCADE;  
DROP TABLE IF EXISTS employee CASCADE;  
DROP TABLE IF EXISTS project CASCADE;  
DROP TABLE IF EXISTS department_locations CASCADE;  
DROP TABLE IF EXISTS department CASCADE;
```

Table Creation

```
-- =====  
-- TABLE: DEPARTMENT  
-- =====  
  
CREATE TABLE department (  
    dnumber INTEGER PRIMARY KEY,  
    dname VARCHAR(50) NOT NULL UNIQUE,  
    mgr_ssn CHAR(9),  
    mgr_start_date DATE,  
    number_of_employees INTEGER DEFAULT 0  
);
```

Update

```
-- Department manager is an employee  
ALTER TABLE department  
ADD CONSTRAINT fk_department_manager  
FOREIGN KEY (mgr_ssn)  
REFERENCES employee(ssn);
```

```
-- =====  
-- TABLE: EMPLOYEE  
-- =====  
  
CREATE TABLE employee (  
    ssn CHAR(9) PRIMARY KEY,  
    fname VARCHAR(30) NOT NULL,  
    minit CHAR(1),  
    lname VARCHAR(30) NOT NULL,  
    bdate DATE,  
    address VARCHAR(150),  
    sex CHAR(1) CHECK (sex IN ('M', 'F')),  
    salary NUMERIC(10,2) CHECK (salary >= 0),  
    super_ssn CHAR(9),  
    dno INTEGER NOT NULL,  
  
    CONSTRAINT fk_employee_department  
        FOREIGN KEY (dno)  
        REFERENCES department(dnumber),  
  
    CONSTRAINT fk_employee_supervisor  
        FOREIGN KEY (super_ssn)  
        REFERENCES employee(ssn)  
);
```

Table Creation

```
-- =====
-- TABLE: DEPARTMENT_LOCATIONS
-- Multivalued attribute: Locations
-- =====

CREATE TABLE department_locations (
    dnumber INTEGER NOT NULL,
    dlocation VARCHAR(50) NOT NULL,

    PRIMARY KEY (dnumber, dlocation),

    CONSTRAINT fk_department_locations_department
        FOREIGN KEY (dnumber)
        REFERENCES department(dnumber)
        ON DELETE CASCADE
);
```

```
-- =====
-- TABLE: PROJECT
-- =====

CREATE TABLE project (
    pnumber INTEGER PRIMARY KEY,
    pname VARCHAR(50) NOT NULL UNIQUE,
    plocation VARCHAR(50),
    dnum INTEGER NOT NULL,

    CONSTRAINT fk_project_department
        FOREIGN KEY (dnum)
        REFERENCES department(dnumber)
);
```

Table Creation

```
-- =====
-- TABLE: WORKS_ON
-- M:N relationship between EMPLOYEE and PROJECT
-- Attribute: Hours
-- =====

CREATE TABLE works_on (
    essn CHAR(9) NOT NULL,
    pno INTEGER NOT NULL,
    hours NUMERIC(4,1) CHECK (hours >= 0),

    PRIMARY KEY (essn, pno),

    CONSTRAINT fk_works_on_employee
        FOREIGN KEY (essn)
        REFERENCES employee(ssn)
        ON DELETE CASCADE,

    CONSTRAINT fk_works_on_project
        FOREIGN KEY (pno)
        REFERENCES project(pnumber)
        ON DELETE CASCADE
);
```

```
-- =====
-- TABLE: DEPENDENT
-- Weak entity dependent on EMPLOYEE
-- =====

CREATE TABLE dependent (
    essn CHAR(9) NOT NULL,
    dependent_name VARCHAR(50) NOT NULL,
    sex CHAR(1) CHECK (sex IN ('M', 'F')),
    birth_date DATE,
    relationship VARCHAR(30),

    PRIMARY KEY (essn, dependent_name),

    CONSTRAINT fk_dependent_employee
        FOREIGN KEY (essn)
        REFERENCES employee(ssn)
        ON DELETE CASCADE
);
```

```
(1, 'Executive Office'),
(2, 'Research and Development'),
(3, 'Software Engineering')
```

NAME: _____

```
(1, 'İstanbul'),
(1, 'London'),
(2, 'İstanbul'),
(2, 'Berlin'),
(3, 'Ankara'),
(3, 'Atina')
```

```

-- DEPENDENTS

```

```
INSERT INTO project (pnumber, pname, plocation, dnum)
VALUES
  (2001, 'Anatolia AI Platform', 'İstanbul', 4),
  (2002, 'European Cloud Migration', 'Berlin', 6),
  (2003, 'Secure Banking Gateway', 'London', 5),
  (2004, 'Multilingual E-Commerce System', 'İstanbul', 3),
  (2005, 'Customer Success Portal', 'Madrid', 12)
```


DATA INSERTION

Please download following file to insert the data into the tables you created.

<https://gazi-end-muh.github.io/dbms/slides/week14.sql>

You must copy and paste the file in the sql studio and run it once.

VERIFY RECORDS

1st: Let's count total number of record on every table with the following code:

```
-- =====  
-- VERIFY COUNTS  
-- =====  
  
SELECT 'department' AS table_name, COUNT(*) AS record_count FROM department  
UNION ALL  
SELECT 'employee', COUNT(*) FROM employee  
UNION ALL  
SELECT 'department_locations', COUNT(*) FROM department_locations  
UNION ALL  
SELECT 'project', COUNT(*) FROM project  
UNION ALL  
SELECT 'works_on', COUNT(*) FROM works_on  
UNION ALL  
SELECT 'dependent', COUNT(*) FROM dependent;
```

Now you need to calculate your attendance

Download and insert the data from following address.

<https://gazi-end-muh.github.io/dbms/slides/week14-attendance.sql>

THE END



CHECK OUT COURSE WEBSITE: [HTTPS://GAZI-END-MUH.GITHUB.IO](https://GAZI-END-MUH.GITHUB.IO)

GOT ANY QUESTIONS LATER?

SEND ME AN EMAIL: DR.ERCAN.EZIN@GMAIL.COM